

What were the highlights last year

One of the major trends that vendors spoke about last year was of application-specific systems on chips (SoCs). There were several versions mentioned like those for consumer electronics (smart TVs or 3D gaming devices) as well as for automotive multimedia systems. These were believed to help reduce the time-to-market for those systems. There were also variants for critical applications like weapon-mounted sights, handheld range and target finders as well as unmanned platforms. HDMI transceivers was another area that was covered, where the audio was ripped off from the video and compressed separately, providing greater flexibility to users.

An article published last year discussed how these solutions sought to tackle the challenges that lie in designing audio systems. Three critical areas that designers seek to maximise are audibility, intelligibility and fidelity. Different applications also bring in their own challenges, such as how high-fidelity sounds are paramount in the audio industry and low data rates have a higher priority in communication systems. One important point from last year was that, if you wanted the best-possible quality, you should go for analogue-processing hardware. This was because a digitally-processed analogue output lost or distorted the signals.

For video, we had looked at how new video-processing applications are growing exponentially, which has created a requirement for computationally demanding video processors that have to be designed into small form factor devices—signs of mobile devices coming out with ultra-high-definition (UHD) screens.

The article also covered new application areas such as automotive systems where vehicle-reversing cameras and small-screen LCDs were driving demand for automotive-grade components and designs, while maintaining low power consumption.

What chips do the hottest consumer devices run on

iPad Air's 5MP rear camera is powered by OmniVision OV2C28BB CIS, while its 1.2MP front camera is powered by OmniVision OV2E0BNN CIS. Its audio codec is Dialog Semiconductor-manufactured yet runs on an Apple-branded chip of model number Apple 343S0655-A1. No other information is available about it.

Own a Samsung Galaxy S6? It has Maxim MAX98505 class DG audio amplifier and Maxim MAX77843 companion PMIC, along with Wolfson WM1840 audio codec. Wolfson is a fabless semiconductor design company based in the UK.

IT-K graduate Sahil S. Patil's Austin based Cirrus Logic has two design wins in Apple's iPhone 5—both audio chips. While Apple 338S1077 is believed to be an encoder, the other is Apple 338S1117 Cirrus audio chip whose function is still unknown.

stereo differential microphones. Its application gets extended to personal media players, smartphones, personal navigation devices, stereo Bluetooth headsets and more.

Advanced headphone solutions are offered by companies like Analog Digital (ADAU1373) and Nuvoton (NAU8822).

The IoT rears its head

With the Internet of Things (IoT) penetrating almost every aspect of electronic design, whether in the name of the IoT or cyber physical systems, traditional manufacturers seem to be getting the proverbial short end of the stick. "Traditional audio/video device manufacturers, who are experts in their fields, may

have difficulty mastering incorporating wireless connectivity and IoT elements in their end products," says Esmond Wong, vice president - business development and supplier marketing, Arrow Electronics.

So, how are vendors tackling this? Advanced Wi-Fi and streaming solutions deliver CD-quality music to anywhere in the premise without the need of connecting wires. Such technologies reproduce the fullness, clarity and depth of music way beyond Bluetooth A2DP.

In addition to longer ranges; higher bandwidth and stronger immunity to interferences, Wi-Fi technologies coupled with UPnP/DLNA, industry standards for music discovery and sharing enable speakers to stream

music from libraries that are typically spread across all compatible devices in the local network, such as PCs, tablets, smartphones and network storage devices.

For instance, you have TI's CC3200 board, which is a microcontroller unit that comes Wi-Fi-certified out of the box. TI has made available audio-booster chips for this board that contain a class D power amplifier to drive the speakers and an ultra-low-power audio codec, TLV320AIC3254, which supports programmable audio processing.

Microchip Technology recently announced, at Hong Kong Electronics Fair, the release of its second-generation Multizone audio technology and a mobile app for use in whole-home-audio and multi-room applications. JB Multizone 2.0 utilises advanced technology to reduce audio drops found in traditional Wi-Fi-streaming applications.

Wi-Fi Direct eliminates the need for each speaker in a zone to connect with the access point, thereby eliminating common problems associated with overloading access points. Bypassing the access point also reduces the radio frequency (RF) bandwidth consumed by half, resulting in increased streaming quality and/or the ability to add more speakers in a zone. The addition of Wi-Fi multimedia (WMM) to Wi-Fi Direct reduces source-to-zone speaker latency and packet loss, resulting in tightly-synchronised audio playback with minimal audio drops.

It is a Wrap

The show goes on, and we will probably see the usual enhancements in audio performance and efficiency next year. There is also a great deal of focus on improving the power efficiency of the chip, while eliminating potential issues with electromagnetic interference, RF interference and audio artefact such as clicks and pops. This year's single-chip solutions as well as comparatively smaller silicon-handling noise-cancellation techniques have also been interesting. ●